CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 1: Schema Definition, Integrity Constraints & DDL

Course: Advanced Database Engineering Laboratory [23CS3506]
Category: Experiments / Practicals | Experiment #1 | Academic Year: 2026-27

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1. Overview & Objective:

Formulate normalized university database schemas enforcing primary keys and foreign key cascades.

2. Practical Tasks & Verification Steps:

1. Create PostgreSQL tables for Department, Faculty, Student, Course, and Enrollment entities.
2. Enforce domain integrity with CHECK constraints e.g. CGPA BETWEEN 0.0 AND 10.0, semester IN 1..8 .
3. Configure ON DELETE CASCADE and ON UPDATE CASCADE foreign key referential actions.
4. Test constraint violation rejections by inserting duplicate primary keys and out-of-range values.

Submission Guidelines & Verification:

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